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Find period of electron:

Centripetal force = magnetic force

$$m\omega v = Bev$$

$$m\left(\frac{2\pi}{T}\right)v = Bev$$

$$\begin{aligned} T &= \frac{2\pi m}{Be} \\ &= \frac{2\pi \times 9.11 \times 10^{-31}}{3.0 \times 1.60 \times 10^{-19}} \\ &= 1.192 \times 10^{-11} \text{ s} \end{aligned}$$

But the electron only travels half a circle, so time spent = $T/2 = 6.0 \times 10^{-12} \text{ s}$.